

YEAR 7 · UNIT 2 · HOMEWORK 6

Letters and Particles

Mathematics 2.1–2.2 — expressions, formulae and substituting

Science 2.1–2.2 — states of matter, particles, changes of state

Section C — the one thing both subjects measured this week

Name: _____

Class: _____

Date given: _____

Date due: **Monday 14 September 2026****How to do this homework**Three sections. Maths first, then science, then **Section C**, where the two meet.

- In maths, **write the middle line** — put the times signs back in first. And remember that $c - 50$ is a **finished answer**: you are allowed to stop.
- In science, answer in **full English sentences**. “Condensation” is not a sentence; “This is condensation, from gas to liquid” is.

Section A · Mathematics 2.1–2.2 — Expressions and Formulae**A1 Expression, formula, or neither?**

Tick one box in each row. The first one is done for you.

The two definitions, side by side

An **expression** contains letters and sometimes numbers, and has **no equals sign**. For example: $n + 7$, $4m$, $x - 3$.

A **formula** is a rule connecting two or more quantities. It contains letters and **has an equals sign**. For example: $A = lw$, $y = 5x$.

	Expression	Formula	Neither
$6h$	X		
$P = 4s$			
$8 + 5 = 13$			
$t - 9$			
$V = lwh$			
$m = 4 + n$			

A2 Write the expression

Mr Bowen has a beaker holding w grams of salt. Write an expression for the mass of salt in each of these.

Word help – the phrases, and the two that decide the operation

more than · fewer than	add · subtract
times as many / as much	multiply
half as much	divide by 2
the total of	add them together
the difference between	subtract the smaller from the larger

Write $3 \times s$ as **3s**, and $s \div 2$ as **s over 2**. The number always goes **in front** of the letter – never $s3$.

- (a) 30 grams more than Mr Bowen _____
- (b) Four times as much as Mr Bowen _____
- (c) 25 grams fewer than Mr Bowen _____
- (d) Half as much as Mr Bowen _____

Now two beakers. The first holds p grams and the second holds q grams. Write an expression for:

- (e) the **total** mass of salt in the two beakers _____
- (f) the **difference** between the two masses _____
- (g) the mass in **six** beakers like the first and **two** like the second _____

A3 The word that changes the order

Word help – read these twice before you start

subtract 6	you take 6 away from what you have: $4x - 6$
subtract from 6	you start at 6 and take away what you have: $6 - 4x$
h less than t	you start at t : $t - h$. The English gives the two letters in the opposite order to the calculation.
the result	the thing you just worked out on the line before

Write an expression for each description.

- (a) Multiply x by 7, then subtract 3. _____
- (b) Multiply x by 7, then subtract the result from 3. _____

(c) g less than k _____(d) y more than nine times z _____(e) Multiply n by 4, then subtract the result from 30. _____

(f) Mr Bowen looks at the expression $9 - 2m$ and says: "Multiply m by 2, then subtract 9."

Is he right? If he is wrong, write the correct description.

A4 Substitute — and put the times sign back in

The middle line

$4n$ was only ever a short way of writing $4 \times n$. So when $n = 3$:

$$4n = 4 \times 3 = 12 \quad \text{not } 43.$$

The middle line is where the marks are. **Write it every time.**

Work out the value of each expression when $n = 6$. Write the middle line.

(a) $n + 8 =$ _____

(c) $n - 11 =$ _____

(b) $5n =$ _____

(d) $\frac{n}{2} =$ _____

A5 The order of operations does not switch off

Times and divide happen first

Even after you have put a number in place of a letter, **times and divide** still happen **before** add and subtract.

$$3x + 2 \quad \text{when } x = 4 \quad \text{is} \quad 3 \times 4 + 2 = 12 + 2 = 14, \quad \text{not } 18.$$

Work these out. **Show the middle line** — an answer with no working cannot be checked.

(a) $6x - 5$ when $x = 4$ _____

(b) $30 - 4n$ when $n = 7$ _____

(c) $2a + 3b$ when $a = 5$ and $b = 4$ _____

(d) $\frac{k}{3} + 9$ when $k = 24$ _____

A6 The minus sign travels with its number**Use brackets**

When the number you are substituting is **negative**, put it in **brackets** so its minus sign cannot be left behind.

$$4 - 3n \text{ when } n = -2: \quad 3n = 3 \times (-2) = -6, \text{ so } 4 - (-6) = 4 + 6 = 10.$$

Work these out when $m = -5$.

(a) $2m =$ _____

(c) $1 - 2m =$ _____

(b) $8 - m =$ _____

A7 Use a formula

Mr Bowen heats a beaker of water. He finds that the temperature in degrees Celsius is given by the formula

$$T = 22 + 6t, \text{ where } t \text{ is the number of minutes since he lit the burner.}$$

(a) Find T when $t = 5$. _____

(b) Find T when $t = 11$. _____

(c) Mr Bowen puts $t = 40$ into the formula and gets $T = 262$. His arithmetic is correct. Explain, using what you know from **Science 2.2**, why the water will never reach that temperature.

A8 Now you write the question**Read all four conditions before you start**

Here is an expression:

$$40 - 5n$$

Write a **word problem** that this expression is the answer to. Your question must:

1. be about something **real** — a beaker, money, a journey, a class;
2. say clearly what **n represents**;
3. **start** at 40, and take away 5 lots of something;
4. be written in **full sentences** and **end in a question**.

Section B · Science 2.1–2.2 – Particles and Changes of State

B1 Match the word to its meaning

Write the correct **letter** in each box.

Word		Meaning
1. Matter	☐	A A space with no particles in it at all.
2. Property	☐	B Squashed into a smaller space.
3. Particle	☐	C The amount of space something takes up.
4. Volume	☐	D Everything you can see and feel.
5. Compressed	☐	E A tiny piece of matter, far too small to see.
6. Vibrate	☐	F A way that a substance behaves.
7. Vacuum	☐	G The curved surface of a liquid in a tube.
8. Meniscus	☐	H To make tiny movements without leaving your place.

B2 The properties table

Put a **tick** for yes and a **cross** for no.

	Solid	Liquid	Gas
Keeps its own shape			
Keeps the same volume			
Can be poured			
Can be compressed			

(b) Dry sand can be poured from one hand to the other. Sand is **not** a liquid. Explain why not.

B3 Answer in full sentences

Sentence starters – you may copy these

“In a solid the particles are...”
“...between the particles...”

“This means that...”

“...because the gaps be-

(a) You can pour water, but you cannot pour a brick. Use **particles** to explain the difference.

(b) Mr Bowen blocks the end of two syringes with his thumb. One is full of **air** and one is full of **water**. Only the air syringe moves when he pushes. Explain why.

(c) A kitchen sponge is a solid, but you can squash it in one hand. Does this prove that particle theory is **wrong**? Explain your answer.

B4 The five changes of state

Complete the table. The first row is done for you.

Doing word	Naming word	From ... to ...	Heat or cool?
melt	melting	solid to liquid	heat
freeze			
	evaporation		
boil			
	condensation		

(b) Write the two temperatures for water:

melting point _____ boiling point _____

B5 Evaporating or boiling?**Same journey, different word**

Evaporation and **boiling** are both liquid to gas. Evaporation is **slow**, happens at the **surface** only, and at **any** temperature. Boiling is **fast**, happens **all through** the liquid, and only at the **boiling point**.

Tick one box in each row.

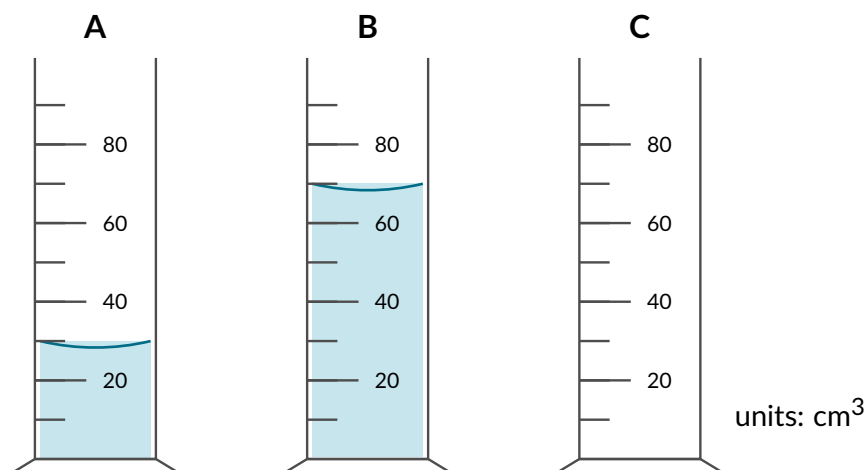
	Evaporating	Boiling
Washing dries on the line on a cool day		
Bubbles form all through the water in a pan		
A puddle disappears from the playground		
It only happens at 100 °C		

(b) Mr Bowen leaves a glass of water on his desk on Friday. On Monday the glass is empty. The water never reached 100 °C. Name the change of state and explain, in full sentences, where the water went.

B6 Read the scale, then draw one**Both instruments, one rule**

Read at **eye level**. On a measuring cylinder, read the **bottom of the meniscus**. On a thermometer, read the **top** of the liquid.

Each small line on the cylinders is 10 cm³. Write the volume in **A** and **B**, then **draw** the water level in **C**.



A = _____

B = _____

C: draw 45 cm³

(b) Two students read cylinder **A** and get two different answers. Neither of them misread the numbers. Explain what went wrong.

Section C · Where the two subjects meet – Volume

In Science you measured a volume in a cylinder. In Maths you built one out of letters. They are the same thing, and this section is why.

The small number counts the lengths

A **length** is one measurement: cm.

An **area** is a length **times** a length: $\text{cm} \times \text{cm} = \text{cm}^2$. Two lengths, so a small **2**.

A **volume** is a length **times** a length **times** a length: $\text{cm} \times \text{cm} \times \text{cm} = \text{cm}^3$. Three lengths, so a small **3**.

So for a box: $V = lwh$ — three letters multiplied, three lengths, cm^3 .

And the unit that joins the two subjects: $1 \text{ cm}^3 = 1 \text{ ml}$, $1000 \text{ cm}^3 = 1 \text{ litre}$.

C1 Count the letters

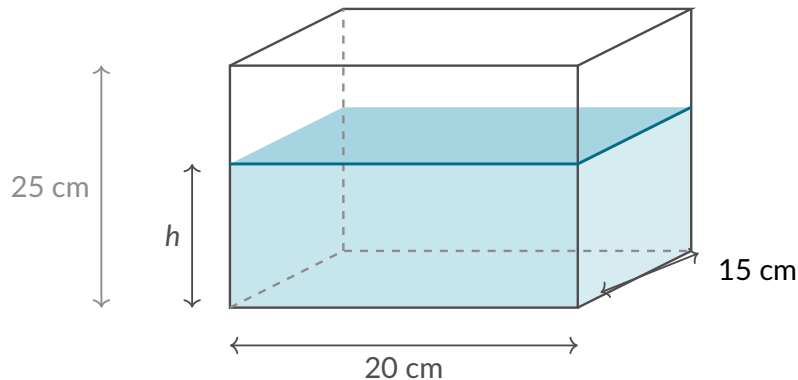
Every letter below stands for a length in centimetres. Count how many lengths are **multiplied together**, then write the unit. The first one is done for you.

Expression	How many lengths multiplied?	Unit
h	one	cm
lw		
lwh		
s^3		
$2l + 2w$		

(b) The last row is different from all the others. Explain why the unit for $2l + 2w$ is **not** cm^2 , even though it has two letters in it.

C2 The tank

A tank has a rectangular base measuring **20 cm** by **15 cm**. It is **25 cm** tall. Mr Bowen pours in water to a depth of h centimetres.



- Write an **expression** for the volume of the water in the tank. _____
- Write it as a **formula** beginning $V =$ _____
- Find the volume of water when $h = 4$. Show the middle line. _____
- How many **litres** of water is that? _____
- Mr Bowen substitutes $h = 40$ into his formula and gets $12\,000\text{ cm}^3$. His arithmetic is correct. Explain why the answer is still wrong.

C3 The same water, three containers

Mr Bowen pours the **same** 150 cm^3 of water from a tall narrow cylinder into a wide flat dish, and then into a beaker.

- As the water moves from container to container, which of the three measurements – its **length**, its **width**, its **height** – change? _____
- What is the value of lwh in every one of the three containers? _____
- Complete the science sentence:
A liquid keeps the same _____ but takes the _____ of its container.
- So l , w and h all change, and lwh does not. Explain in your own words how those two facts fit together.

C4 The syringe — where the volume really does change

A sealed syringe holds $v \text{ cm}^3$ of **air**. Nothing can get in or out.

- (a) Mr Bowen pushes the plunger until the air fills **half** the space it did before. Write an expression for the new volume of the air. _____
- (b) Mr Bowen empties the syringe, fills it with $v \text{ cm}^3$ of **water** instead, seals it, and pushes just as hard. Write an expression for the new volume of the water, and say why.
- _____
- _____
- _____

C5 Three bad ideas

Mr Bowen has had three ideas. Each one is a bad idea. For each, say **what goes wrong**, in a full sentence, using the science or the maths from this unit.

1. He wants a smaller cup of tea, so he seals the cup and squeezes it very hard.
- _____
- _____
2. His formula says the water will be at 262°C after 40 minutes, so he plans to leave the burner on and come back at lunchtime.
- _____
- _____
3. He measures a box as 10 cm long, 10 cm wide and 10 cm tall, and writes that its volume is 30 cm^3 .
- _____
- _____
- _____

Answer Key – Teacher Copy

A1 Expression, formula, or neither?

$6h$ – expression $P = 4s$ – formula $8 + 5 = 13$ – neither $t - 9$ – expression
 $V = lwh$ – formula $m = 4 + n$ – formula

The row that separates the class is $8 + 5 = 13$. It has an equals sign, so students who learned “equals sign means formula” tick formula – but it has no letters, so it is neither. Both halves of each definition have to be checked, and this row is the only place that shows who is checking one.

A2 Write the expression

(a) $w + 30$ (b) $4w$ (c) $w - 25$ (d) $\frac{w}{2}$ (e) $p + q$ (f) $p - q$ (g) $6p + 2q$

(b) is the one to watch: $w4$ instead of $4w$. The number goes in front, always, and it is worth correcting on sight rather than in writing.

(f) has no single right answer and that is fine – $q - p$ is equally good, as long as the student can say which beaker they started at. What is *not* acceptable is $p + q$, which means the word *difference* was read as “the answer”.

(g) is the two-letter question. A student who writes $8pq$ has merged two different quantities into one and needs the beakers drawn.

A3 The word that changes the order

(a) $7x - 3$ (b) $3 - 7x$ (c) $k - g$ (d) $9z + y$ (e) $30 - 4n$

(f) He is **wrong**. His description gives $2m - 9$. The expression $9 - 2m$ starts at 9, so the description needs *subtract the result from 9*.

(a) and (b) are the same five words with one preposition changed, and they are printed next to each other on purpose. If a student gives the same answer to both, they read the numbers and not the sentence – do not mark it, make them read both aloud.

(c) is $t - h$ with new letters. Check it with numbers if anyone resists: 5 less than 12 is 7, and nobody writes $5 - 12$ for that.

(f) is the best question in Section A after A8, because it runs the translation *backwards*. A student who cannot spot the error can still often fix it once told there is one – that is worth knowing, and it is the difference between not reading and not being able to.

A4 Substitute

(a) $6 + 8 = 14$ (b) $5 \times 6 = 30$ (c) $6 - 11 = -5$ (d) $6 \div 2 = 3$

(b) is the headline error of the whole unit: **56**, two digits pushed together, written with total confidence by a student whose arithmetic is perfect. It is our own fault – in 2.1 we taught them to write $5 \times n$ as $5n$ – so treat it as a reading error and put the times sign back in on the board rather than marking the arithmetic.

(c) goes below zero, which is Unit 1 work and is allowed. A student who writes 5 has decided an answer cannot be negative.

A5 The order of operations

- (a) $6 \times 4 - 5 = 24 - 5 = 19$
 (b) $30 - 4 \times 7 = 30 - 28 = 2$
 (c) $2 \times 5 + 3 \times 4 = 10 + 12 = 22$
 (d) $24 \div 3 + 9 = 8 + 9 = 17$

(b) is the one to slow down on. The multiplication is written *second* and still happens first; a student working left to right gets $26 \times 7 = 182$. This is also the shape of C2(e) and of every “start at 40 and take away” question in the unit.

Mark the middle line, not the answer. Every error on this question is invisible in a final number and obvious in the working, which is the entire argument for demanding it.

A6 The minus sign travels

- (a) $2 \times (-5) = -10$ (b) $8 - (-5) = 8 + 5 = 13$ (c) $1 - 2 \times (-5) = 1 - (-10) = 1 + 10 = 11$

(c) is the question. The wrong answer is -9 , from substituting 5 and leaving the minus sign behind. Insist on the brackets at the substitution step; that one habit fixes the whole class of error.

Note that the arithmetic in (c) is $1 + 10$, and nobody in this room finds that hard. Say so out loud — the difficulty is entirely in getting the sign into the expression in one piece.

A7 Use a formula

- (a) $T = 22 + 6 \times 5 = 22 + 30 = 52^\circ\text{C}$ (b) $T = 22 + 6 \times 11 = 22 + 66 = 88^\circ\text{C}$

(c) The water **boils at 100°C and stays there**. The formula describes the water heating up; once it reaches its boiling point that situation has ended, so the formula has ended with it.

(c) **is the most valuable question in Section A**. The arithmetic is perfect and the answer is still wrong. If a student says the water would have boiled away, agree warmly — they are right, and they have made the point for you. Anyone who answers “he did the sum wrong” has not read the sentence saying he did not.

In (a) and (b), watch for the 22 being multiplied: $6t$ with $t = 5$ is 30, and the 22 is added once, not six times. It is A5’s error in a formula.

A8 Now you write the question

No single right answer. Check the four conditions **in this order**:

1. **Does the story start at 40?** It must *begin* there, not arrive there. This is the condition most of them will miss.
2. **Is it 5 lots of something, repeated?** Five taken away once is not $5n$ — it needs a rate: 5 each day, 5 per person, 5 grams every minute.
3. **Is it said what n represents?**
4. **Full sentences, ending in a question?**

One that works: “A beaker holds 40 grams of salt. Mr Bowen takes out 5 grams every minute. Write an expression for the mass of salt left after n minutes.”

A question that ends “what is $40 - 5n$?” has not done the task — it is the expression copied out in words. Name that out loud rather than marking it wrong.

This question cannot be answered by pattern-matching, which makes it the single best thing on the packet for finding out who is reading. Read four or five of them to the class.

B1 Match the word to its meaning

1-D 2-F 3-E 4-C 5-B 6-H 7-A 8-G

Property (2-F) is the one that goes wrong, and not because of the science: the everyday English meaning is a house or land, and a student reaching for that meaning has nowhere to put F.

Vibrate (6-H) is the other: it means small movements **on the spot**, not travelling. Ask a student to vibrate in their chair without leaving it; it settles in five seconds.

B2 The properties table

	Solid	Liquid	Gas
Keeps its own shape	yes	no	no
Keeps the same volume	yes	yes	no
Can be poured	no	yes	yes
Can be compressed	no	no	yes

(b) Every **grain** of sand keeps its own shape and cannot be compressed, so every grain is a solid. What flows is not one substance pouring — it is millions of tiny solids rolling over one another.

The liquid column is the useful one: it is the only column with a *yes* and a *no* in the middle two rows, and that pair is exactly what Section C is built on. Point at it when you mark C3.

(b) is a full-sentence question and the sentence matters. “Because it is a solid” is circular; push for the grain.

B3 Answer in full sentences

(a) Matter can only flow if the particles can **move past one another**. In a solid the attractive forces are strong and hold every particle in its place. In a liquid the forces are weak enough to let particles slide past each other, but still strong enough to keep them touching.

(b) A gas has **large gaps** between its particles, so pushing simply moves them closer together and the air takes up less room. In water the particles are already touching, so there is no gap left to close and the plunger will not move. Nothing escapes from either syringe.

(c) **No**. The sponge is not solid all the way through — it is a thin solid skeleton with **air** in every hole. Squeezing it compresses the air, not the solid, and the air pushes it back when you let go.

(c) is the best question in Section B and should be discussed rather than marked. It is the first time this class has been asked to *judge* a model instead of learn one, and a student who says “the theory is wrong” has done something intellectually respectable — they have taken the evidence seriously. Give them the sponge and let them find the air.

In (b), watch for “the air escaped”. Nothing escaped; his thumb is on the end. That answer means the gaps have not landed.

B4 The five changes of state

melt	melting	solid to liquid	heat
freeze	freezing	liquid to solid	cool
evaporate	evaporation	liquid to gas	heat
boil	boiling	liquid to gas	heat
condense	condensation	gas to liquid	cool

(b) Melting point **0 °C** Boiling point **100 °C**.

The two rows in the middle are the same journey with different words, and a student who notices that and asks about it has understood the section.

The spellings are the marks here: *evaporation*, not “evaporating”; and *condensation*, not “condensing”. Both irregular nouns are worth saying aloud with the class, twice.

B5 Evaporating or boiling?

Washing on the line — **evaporating** Bubbles all through the pan — **boiling**

A puddle disappears — **evaporating** Only at 100 °C — **boiling**

(b) **Evaporation.** The liquid water turned slowly into **water vapour**, an invisible gas, at the surface, at room temperature, and spread out into the air of the room.

Row 1 is the row that decides it. “Cool day” is in the sentence precisely so that a student who believes liquid-to-gas requires heat has to face it.

In (b), “it disappeared” and “it dried up” are descriptions, not answers. The answer needs the word *water vapour* and the word *gas*. This is the same question as the puddle from the lesson, and it should come back better than it did in class.

B6 Read the scale, then draw one

A = 30 cm³ B = 70 cm³ C — the line should be drawn **halfway between the 40 and the 50 marks**.

(b) They looked from **different angles**. Looking down onto the meniscus from above gives a reading that is too high, and looking up from below gives one that is too low. The fix is to read the **bottom of the meniscus at eye level**.

C is the question, and it is the reverse skill. Reading a scale can be done by pattern; putting a line at 45 cannot, because there is no line there to copy — the student has to know that each gap is 10 and that 45 is halfway into one. Expect some lines drawn on the 40 or on the 50.

Mark down any answer to A or B with no unit. This is the third packet in which units have been asked for, and it should now be automatic.

C1 Count the letters

<i>h</i>	one	cm
<i>lw</i>	two	cm ²
<i>lwh</i>	three	cm ³
<i>s</i> ³	three (it is $s \times s \times s$)	cm ³
$2l + 2w$	none — they are added	cm

(b) Because the lengths are **added**, not multiplied. $2l + 2w$ is four lengths laid end to end — it is the distance all the way round the edge, which is still a length.

The last row is the whole question and most of the class will get it wrong, because the rule they will have extracted is “count the letters” and here there are two letters and the answer is still cm. That is the useful failure: the rule is count the lengths *multiplied*. Do not mark this one — put both answers on the board and let the class argue, then walk a finger round the edge of a rectangle while saying “length, length, length, length”.

s^3 catches students who count *letters* rather than lengths and answer cm. The little 3 is already telling them; that is the point of the section.

C2 The tank

- (a) $20 \times 15 \times h$, which is **300h** (b) **$V = 300h$**
 (c) $V = 300 \times 4 = \mathbf{1200\text{ cm}^3}$ (d) $1200 \div 1000 = \mathbf{1.2}$ litres
 (e) The tank is only **25 cm tall**, so the water cannot be 40 cm deep. The formula describes water *inside this tank*; at $h = 40$ that situation no longer exists, so the formula has stopped being true. (The most water the tank can hold is $300 \times 25 = 7500\text{ cm}^3$.)
 (a) is the wall from 2.1 in a new coat. $300h$ does not equal anything, and a student who leaves it blank or invents a number has not accepted that an expression is a finished answer. Say it again: *you are allowed to stop*.
 (c) is the wall from 2.2 in the same coat. $300h$ with $h = 4$ is 1200, never **3004**. If both errors appear in one student's work, the two lessons have been learned separately and not joined up – which is exactly what this question was built to find out.
 (e) is A7.2(c) again with a different quantity, and asking it twice in one packet is deliberate. A formula is a rule about a situation. When the situation ends, so does the rule.

C3 The same water, three containers

- (a) **All three** can change – the water is longer and wider and shallower in the dish than in the cylinder.
 (b) $lwh = \mathbf{150\text{ cm}^3}$, in every container. It does not change.
 (c) A liquid keeps the same **volume** but takes the **shape** of its container.
 (d) The three measurements are trading against each other. Making l and w bigger forces h to get smaller by exactly as much, so the product stays at 150.

This is the question the packet was built around. It is the science property from 2.1a – one sentence the class has already copied down – and it is also an algebraic fact: three letters all change and their product does not. Neither subject could have said this on its own.

(d) is a stretch and a short answer is fine. “The height goes down when the width goes up” is worth full credit. What you are listening for is whether the student sees the *trade*; anyone who writes “because it is the same water” has the right instinct and should be pushed one step further with the dish in front of them.

If the class struggles, pour 150 cm^3 between a cylinder and a dish at the front. It takes twenty seconds and it settles the whole question.

C4 The syringe

- (a) $\frac{v}{2}$ (accept $v \div 2$)
 (b) The volume is still **v**. It does not change at all, because water cannot be compressed – its particles are already touching, so there is no gap to close.
(b) is the question worth the most and it will be left blank by some. The answer is a single letter and no operation, and students do not believe that “write an expression” can be answered with the letter they were given. It can. This is the same discomfort as $c - 50$ in 2.1, arriving from the science side. Note what the pair actually shows: the same push, the same syringe, and the volume halves for one substance and does not move for the other. The maths did not decide that. The particles did.

C5 Three bad ideas

- Tea is a **liquid**, and a liquid cannot be compressed – its particles are already touching, so there is no gap to close. Squeezing will not make the tea smaller however hard he pushes.
- Water **boils at 100°C and stays there**. The formula stops being true at the boiling point, so the water will never reach 262°C ; it will boil, and if he leaves it until lunchtime it will all have boiled away.
- He **added** the three lengths instead of **multiplying** them. The volume is $10 \times 10 \times 10 = \mathbf{1000\text{ cm}^3}$. The little 3 in cm^3 is the reminder: three lengths, multiplied.

Read these three aloud with a completely straight face. Do not explain that they are jokes; the class finds them funnier without help, and the straight delivery is what makes them argue back.

Each one is a different subject failing: 1 is particle theory, 2 is a formula outliving its situation, 3 is arithmetic dressed as a unit error. A student who gets 3 and misses 1 is not weak at maths — they have not joined the two halves of the week together, which is what Section C exists to fix.

If you mark only one thing on this page, mark 3. It is the last question in the packet and it is the sentence the whole of Section C was written to make obvious.